As figure 1.2 illustrates, you do not need to learn every aspect of these topics to know how to build secure APIs. We will instead pick the most critical areas from each and blend them to give you a thorough understanding of how they apply to securing an API.

From information security you will learn how to

- Define your security goals and identify threats
- Protect your APIs using access control techniques
- Secure information using applied cryptography

> **DEFINITION *Cryptography*** is the science of protecting information so that two or more people can communicate without their messages being read or tampered with by anybody else. It can also be used to protect information written to disk in which case it may be the same person reading it at a later time.

From network security you will learn

- The basic infrastructure used to protect an API on the internet, including firewalls, load-balancers, and reverse proxies, and roles they play in protecting your API (see next section)
- Use of secure communication protocols such as *HTTPS* to protect data transmitted to or from your API

> **DEFINITION *HTTPS*** is the name for HTTP running over a secure connection. While normal HTTP requests and responses are visible to anybody watching the network traffic, HTTPS messages are hidden and protected by Transport Layer Security (TLS, also